ModelIt!

Student Activity Pack

The Water-Energy-Food Nexus: Three Crises, One System

Modeling Coupled Resource Systems Under Population Pressure

Grade 09L2 | NGSS HS-ESS3-4, HS-ETS1-2 | Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 How much water do you think it takes to produce electricity?

2 Why might solving a water shortage cause an energy crisis?

3 Draw a diagram showing how water, energy, and food production might be connected.

4 What happens to all three resources when a city population doubles?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Water Availability | Energy Production | Agricultural Output | Population Demand | Industrial Usage | Wastewater Recycling

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Growing City

Population +30 percent over decade

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Drought

Water -40 percent sudden reduction

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Circular Economy

Wastewater recycling at 90 percent

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Desalination Capacity — Energy-intensive conversion of seawater to freshwater, trading energy for water

Renewable Energy Fraction — Percentage of energy from low-water sources like solar and wind

Urban Agriculture — Food production within city limits using vertical farms and hydroponics

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why does solving one resource crisis often make another worse? Use evidence from your simulation to support your answers.

1 How does your model show that resource crises are interconnected rather than independent?

2 Why does optimizing one resource in isolation often make another worse?

3 What role does Wastewater Recycling play in decoupling the competing systems?

4 How does population growth create compounding rather than linear pressure?

5 What trade-offs did you accept in your sustainable city design?

8 STEM Challenge: Design a Sustainable City Resource Plan

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YOUR ENGINEERING MISSION

Design an integrated water-energy-food management plan for a growing city that avoids resource trade-off traps.

THE SCENARIO

A city of 2 million is expected to grow to 3 million in 15 years. Water supply is already strained, energy costs are rising, and surrounding farmland is being developed. Design a plan that meets all three resource needs sustainably.

Engineering Design Thinking

1. HOW WOULD YOU ALLOCATE WATER BETWEEN ENERGY PRODUCTION, AGRICULTURE, AND HOUSEHOLDS?
2. WHAT MIX OF ENERGY SOURCES MINIMIZES WATER CONSUMPTION?
3. HOW WOULD YOU INTEGRATE WASTEWATER RECYCLING INTO THE FOOD PRODUCTION SYSTEM?
4. HOW WOULD YOUR PLAN ADAPT IF DROUGHT REDUCES WATER SUPPLY BY 30 PERCENT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Environmental Systems Engineers design integrated resource management solutions for cities and regions, earning $65,000-$130,000/year at consulting firms, utilities, and government agencies.